Seon-Hwan Hwang, Ph.D.

Full Professor
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RESEARCH INTERESTS:

Motor Drives for PMSM and BLDC
- High speed single-phase PMSMs, Drives of Multi-phase PMSMs and 3-phase PMSM

Power Converters
- Controls of wind turbine generator and grid-connected power converters
- Control strategies : current controls, PLL, dead-time compensation, power quality, and etc.

Software/Hardware Implementations
- Design of embedded systems and HILS based on DSP

PROFESSIONAL: EXPERIENCE

Kyungnam University, Changwon, Korea

Full Professor **Mar '23 ~**
Associate Professor **Mar '18 to Feb '23**
Assistant Professor **Mar '12 to Feb '18**

- Director of Power Conversion Research Lab.
- Course Taught
  - Undergraduate Courses : Power Electronics, Electronic Circuit, Measurement Engineering, Digital Engineering
  - Graduate Courses : Digital System Control, Advanced Energy Conversion, Analysis of Electric Machines, Advanced Power Electronics

CAPS, Florida State University, FL, USA

Visiting Scholar **Feb '24 to Feb '25**

- Collaborative Research on Power Electronics and Advanced Motor Drives
- Supervisor : Prof. Hui Li

CAPS, Florida State University, FL, USA

Postdoctoral Researcher **Feb '11 to Feb '12**

- Collaborative Research Projects : PV applications using SiC devices, Design of control board(DSP+FPGA), PMSM motor drives
- Supervisor : Prof. Hui Li

EDUCATION:

Pusan National University, Pusan, Korea

Ph.D., Electrical Engineering, **Feb '11**

- Dissertation : A Study of Control Performance of MM-TFPMG for Large-scale Direct-driven Wind Turbines
- Supervisor: Prof. Jang-Mok Kim

M.S., Electrical Engineering, **Feb '06**

- Dissertation : A Dead Time Compensation Method in Voltage-Fed PWM Inverter
- Supervisor: Prof. Jang-Mok Kim

Pusan National University, Pusan, Korea

B.S., Electrical Engineering, **Feb '04**

MEMBERSHIPS:

Institute of Electrical and Electronics Engineers (IEEE), Member, 2012-present
The Korea Institute of Electrical Engineers (KIEE), Member, 2012-present
The Korea Institute of Power Electronics (KIPE), Member, 2012-present
The Korea Institute of Illuminating and Electrical Installation Engineers (KIIEE), Member, 2012-present

SERVICE:

Editor

- Journal of Electrical Engineering & Technology (JEET) – SCI/E indexed

Reviewer Service

- IEEE Transactions on Industrial Electronics
- IEEE Transactions on Power Electronics
- Journal of Power Electronics
- Energies

PUBLICATIONS:

[1] Yongha Choo, Hongsik Hwang, Jeonghyun Cho, Changhwan Kim, JeongMyung Kim, Seon-Hwan Hwang, Ju Yong Choi and Cheewoo Lee, "Investigation of Systematic Efficiency in a High-Speed Single-Phase Brushless DC Motor Using Multi-Physics Analysis for a Vacuum Cleaner", IEEE Trans. on Magnetics, Vol. 55, No. 7, July 2019.
[2] Hongsik Hwang, Jeonghyun Cho, Seon-Hwan Hwang, Ju Yong Choi and Cheewoo Lee, "Design of a single-phase BLDC Motor for a Cordless Vacuum Cleaner Considering the Efficiency of Airflow", Energies, 2019, Vol. 12, No. 3
[3] Jae Suk Lee and Seon-Hwan Hwang, " DC Offset Error Compensation Algorithm for PR Current Control of a Single-Phase Grid-Tied Inverter", Energies 2018, Vol. 11, No. 9
[4] Hae-Jin Kim, Seon-Hwan Hwang and Jang-Mok Kim, "Novel Soft Starting Algorithm of Single Phase Induction Motors by Using PWM Inverter", Journal of Power Electronics, Vol. 18, No. 6, pp. 1720-1728, Nov. 2018.
[5] Seon-Hwan Hwang and Tae-Uk Jung, "Optimal current control method for efficiency improvement of bldc motor", Far East Journal of Electronics and Communications, Vol. 17, No. 6, pp. 1582-1586, Dec. 2017.
[6] Seon-Hwan Hwang and Deok-Je Bang, "Wide Air-gap Control for Multi-module Permanent Magnet Linear Synchronous Motors without Magnetic Levitation Windings", Journal of Power Electronics, Vol. 16, No. 5, pp. 1773-1780, Sept. 2016.
[7] Ui-Seok Seong and Seon-Hwan Hwang, "Analysis of phase error effects due to grid frequency variation of SRF-PLL based on APF", Journal of Power Electronics, Vol. 16, No. 1, pp. 18-26, Jan. 2016.
[8] Dong-Yoon Kim, Won-Sang Lim, Seon-Hwan Hwang and Jang-Mok Kim, "Compensation of Current Offset Error in Half-Bridge PWM Inverter for Linear Compressor", Journal of Power Electronics, Vol. 15, No. 6, pp. 1593-1600, Nov. 2015.
[9] Seon-Hwan Hwang, Dong-Yoon Kim, Do-Hyeon Jang and Jang-Mok Kim Jin-Hyung Yoo, Tae Uk Jung, "Signal Compensation for Analog Rotor Position Errors due to Nonideal Sinusoidal Encoder Signals", Journal of Power Electronics, Vol. 14, No. 1, pp. 82-91, Jan. 2014.
[10] Liming Liu, Hui Li, Seon-Hwan Hwang and Jang-Mok Kim, " An Energy-Efficient Motor Drive With Autonomous Power Regenerative Control System Based on Cascaded Multilevel Inverters and Segmented Energy Storage", IEEE Trans. on Industry Applications, Vol. 49, No. 1, pp. 178-188, Jan./Feb. 2013.
[11] Seon-Hwan Hwang, Xiaohu Liu, Jang-Mok Kim and Hui Li, "Distributed Digital Control of Modular-Based Solid-State Transformer Using DSP+FPGA", IEEE Trans. on Industrial Electronics, Vol. 60, No. 2, pp. 670-680, Feb. 2013.

[12] Seon-Hwan Hwang, Liming Liu, Hui Li and Jang-Mok Kim, "DC Offset Error Compensation for Synchronous Reference Frame PLL in Single-Phase Grid-Connected Converters", IEEE Trans. on Power Electronics, Vol. 27, No. 8, pp. 3467-3471, Aug. 2012.
[13] Je-Wook Park, Seon-Hwan Hwang and Jang-Mok Kim, "Sensorless Control of Brushless DC Motors With Torque Constant Estimation for Home Appliances", IEEE Trans. on Industry Applications, Vol. 48, No. 2, pp. 677-684, Mar. 2012.
[14] Seon-Hwan Hwang, H. J. Kim, J. M. Kim, L. Liu, and H. Li, "Compensation of Amplitude Imbalance and Imperfect Quadrature in Resolver Signals for PMSM Drive," IEEE Trans. Ind. Applicant., vol. , no. , pp. 49-156, Sept./Oct. 2010.
[15] Seon-Hwan Hwang and J. M. Kim, "Dead Time Compensation Method for Voltage-Fed PWM Inverter," IEEE Trans. Energy Conversion, vol. 25, no. 1, pp. 1-10, 2010.
[16] H.S. Jung, Seon-Hwan Hwang, J. M. Kim, C. U. Kim, and C. Choi, "Diminution of Current-Measurement Error for Vector-Controlled AC Motor Drives," IEEE Trans. Ind. Applicant., vol. 42, no. 5, pp. 1249-1256, Sept./Oct. 2006.

INTERNATIONAL: CONFERENCES

[1] Jin-Hyung Yoo, Seon-Hwan Hwang and Tae-Uk Jung, "Stator Teeth Paring Design Optimization of Dual Radial Flux Permanent Magnet Generator for Cogging torque Minimization", 4th North American Conference on Industrial Engineering & Operations Management, Toronto, Canada, Oct. 2019.
[2] Tae-Seong Kim, Seon-Hwan Hwang, Jin-Soo Kim and Jong-Won Park, "DC Offset Compensation Algorithm in the Grid Voltage of single phase Grid-connected Inverter", 2019 IEEE Applied Power Electronics Conference and Exposition (APEC), Mar. 2019.
[3] Hyeong-Do Kang and Seon-Hwan Hwang, "Starting Method of High Speed Operation Sinlge-phase BLDC Motor Based on Virtual DQ Current Control", 2018 21st International Conference on Electrical Machines and Systems (ICEMS), Jeju, Korea, Oct. 2018.
[4] Hyeong-Do Kang, Seon-Hwan Hwang and Jaesuk Lee, "Initial Rotor Position Estimation of Single-phase Permanent Magnet Synchronous Motor with Asymmetric Air-gap", 2018 IEEE Transportation Electrification Conference and Expo, Bangkok, Thailand, Jun. 2018.
[5] Gwang-Hyun Shin, Jae Suk Lee and Seon-Hwan Hwang, "Development of a Speed Ripple Reduction Algorithm for an Oil Cooler Compressor", 2017 IEEE 3rd International Future Energy Electronics Conference and ECCE Asia, Kaohsiung, Taiwan, Jun. 2017.
[6] Kwang-Hyun Shin, Seon-Hwan Hwang and Jae Suk Lee, "Analysis of an Offset Error on a Single-phase Grid-connected Inverter Based on a Proportional-resonant Controller", 2017 IEEE Applied Power Electronics Conference and Exposition (APEC), Tampa, USA, Mar. 2017.
[7] Eun-Soo Kim, Ui-Seok Seong, Jae Suk Lee and Seon-Hwan Hwang, "Compensation of Dead Time Effects in Grid-tied Single-phase Inverter using SOGI", 2017 IEEE Applied Power Electronics Conference and Exposition, Tampa, U.S.A., Mar. 2017.
[8] Chan-Nyeong Heo, Seon-Hwan Hwang and Deok-Je Bang, "Current Ripple Reduction Due to Current Measurement Errors of Independent Three-phase Permanent Magnet Synchronous Machine with Dual Air-gap", 2015 IEEE 2nd International Future Energy Electronics Conference and ECCE Asia, Taipei, Taiwan, Nov. 2015.
[9] Ui-Seok Seong and Seon-Hwan Hwang, "Analysis of Phase Error Effects between Virtual Two-phase Voltages in SRF-PLL Based on APF", 2015 9th

International Conference on Power Electronics and ECCE Asia (ICPE-ECCE Asia), Seoul, Korea, Jun. 2015.
[10] Chan-Nyeong Heo, Byeong-Guk Jeong, Seon-Hwan Hwang and Deok-Je Bang, "Double-sided Air Gap Control of Module-based Permanent Magnet Synchronous Machine for Low-speed and High-torque Applications", 2015 9th International Conference on Power Electronics and ECCE Asia (ICPE-ECCE Asia), Seoul, Korea, Jun. 2015.
[11] Deok-Je Bang, Seon-Hwan Hwang, Ji-Won Kim, Wook Hwang, Pil-Wan Han and Dae-Hyun Koo, "Bearingless Transverse Flux Permanent Magnet Machine for Large Direct-drive", 2014 IEEE Energy Conversion Congress and Exposition (ECCE), Pittsburgh, PA, USA, Sept. 2014.
[12] Seon-Hwan Hwang, Deok-Je Bang and Ji-Won Kim, "Air Gap Control of Multi-phase Transverse Flux Permanent Magnet Linear Synchronous Motor by Using Independent Vector Control", 2014 International Power Electronics Conference, Hiroshima, Japan, May 2014.
[13] Seon-Hwan Hwang, Jang-Mok Kim, Deok-Je Bang, Ji-Won Kim, Dae-Hyun Koo and Do-Hyun Kang "Control of Independent Multi-phase Transverse Flux Linear Synchronous Motor Based on Magnetic Levitation", 2014 IEEE Applied Power Electronics Conference and Exposition, Fort Worth, TX, USA, Mar. 2014.
[14] Deok-Je Bang, G. W. Jang, Seon-Hwan Hwang, Pil-Wan Han, Ji-Won Kim, Do-Hyun Koo, Henk Polinder and J. A. Ferreira, "New Bearingless Generator with Buoyant Rotor for Large Direct-Drive Wind Turbines", 2014 European Wind Energy Association (EWEA), Spain, Mar. 2014.
[15] Dong-Youn Kim, Jang-Mok Kim, Je-Wook Park and Seon-Hwan Hwang, "Compensation of Current Measurement Error in Half-bridge PWM Inverter for Linear Compressor", 2013 IEEE Energy Conversion Congress and Exposition (ECCE-ASIA), Melbourne, VIC, Australia, Jun. 2013.
[16] Jong-Joo Moon, Je-Wook Park, Jang-Mok Kim, Seon-Hwan Hwang and Deok-Je Bang, "Phase current estimation of BLDC motor drive system by using source side current of DC-link", 2012 IEEE Energy Conversion Congress and Exposition (ECCE), Raleigh, NC, USA, Sept. 2012.

PATENTS:

[1] Starting Current Reduction Method and Device of Single-phase Induction Motor Based on dq-axis Transformation, Korea, Registration No. 10-1998515-0000, Jul. 2019.
[2] Power Control Type Energy Management System Using Lithium Ion Battery and Super-capacitor, Korea, Registration No. 10-1863141-0000, May 2018.
[3] Power-controlled Energy Storage Device Using Lithium Battery and Supercapacitor, Korea, Registration No. 10-1863138-0000, May 2018.
[4] Phase Error Compensation Method According to the Grid Frequency Variation of the Single-phase Grid-connected Inverters, Korea, Registration No. 10-1789403-0000, Oct. 2017.
[5] Offset and Scale Error Reduction Method According to Tracing Grid Phase Angle of Three-phase Grid-connected Inverters, Korea, Registration No. 10-1768800-0000, Aug. 2017.
[6] Reduction Method of Current Measurement Errors for High Performance Operation of Three Phase PMSM, Korea, Registration No. 10-1763487-0000, Jul. 2017.